"Freeze the plan before you freeze the code -- architecture, ADRs, and a backlog everyone can defend."

MODULE 48

Capstone Planning and Architecture

Turn the Northstar CRM brief into an executable architecture and delivery plan: C4 context and container views, measurable NFRs, a prioritized vertical backlog, ADRs, and a scored risk register that Labs 49-52 build against.


TEKsystems
Global Services
WORKFORCE
DEVELOPMENT


TEKsystems
Global Services
WORKFORCE
DEVELOPMENT

MODULE 48 OVERVIEW

Produce C4 context and container views, measurable NFRs, a prioritized backlog, ADRs, and a scored risk register -- the plan Labs 49-52 build against.

Week 6 opens with the Northstar CRM capstone: a large enterprise needs a customer management platform for its service agents. Before any code is written, the team must freeze scope, architecture, NFRs, backlog, and risks -- reviewers will not fund disconnected demos.

DURATION & LAB

4 Hours Theory

Architecture views, requirements, NFR writing, backlog and ADR authoring, and risk planning.

2 Hours Lab

lab48-crm: context and container diagrams, NFRs, backlog, five or more ADRs, and a risk register.

Scenario

Northstar CRM for service agents -- Amina (CUS-1001) and Ravi (CUS-1002) as the fixed demo fixtures.

MODULE ARC

Architecture First

C4 context and container views before any backend or frontend code exists.

Vertical Backlog

CAP-12 and friends -- stories that cross API, UI, data, and events, never horizontal layers.

Scored Risk

Every risk gets a likelihood, an impact, a mitigation, and an owner -- never just 'hope'.

KEY TAKEAWAY Total time: 4 hours theory + 2 hours lab = 6 hours. No Lab 49-52 work counts as in scope unless it maps to a backlog item, an ADR, and a measurable NFR.

WHY ENTERPRISE PROJECTS NEED PLANNING

Leadership will not fund disconnected demos -- reviewers need traceability from business outcomes to architecture, stories, tests, and deployment.

Traceability or Rejection

Every Lab 49-52 change must trace to a backlog item, an ADR, or an explicit out-of-scope note.

Ambiguity Blocks Acceptance

Vague 'fast' or 'scalable' language and missing trust boundaries are acceptance-blockers at the Lab 52 defense.

Architecture Before Code

Context and container views, NFRs, and ADRs come before any Spring Boot or React work begins.

Graded Like Code

Lab 48 is scored on the same 100-point rubric rigor as any coding lab in this bootcamp.

KEY TAKEAWAY A peer must be able to reproduce the Week 6 plan from your docs alone -- pretty diagrams without trust boundaries or owners lose real marks.

Why Enterprise Projects Need Planning

MODULE LEARNING OBJECTIVES

By the end of this module, you will be able to plan an enterprise architecture a team can actually build against.

1. Clarify business scope, users, journeys, exclusions, and success measures.
2. Produce C4 context diagrams with protocols and trust boundaries.
3. Design container views placing React, Spring Boot, PostgreSQL, Kafka, identity, and observability.
4. Define domain ownership and versioned HTTP and event contracts.
5. Write measurable NFRs with method, environment, and thresholds.
6. Slice a vertical backlog with acceptance criteria, including story CAP-12.
7. Record architecture decisions as ADRs with alternatives and consequences.
8. Plan ownership, critical path, and a scored risk register with mitigations.

KEY TAKEAWAY These eight objectives are drawn directly from the graded lab -- you will produce all six core artifacts: context, containers, NFRs, backlog, ADRs, and risk register.

Module Learning Objectives

CAPSTONE PROJECT OVERVIEW

Northstar CRM is a customer management platform for service agents -- one coherent product, not five disconnected demos.

Field	Value
Codename	Northstar CRM
Java Package	com.northstar.crm
Repo Path	~/java-bootcamp/examples/customer-management-platform/
Primary Users	Service agent, manager, platform operator

DEMO FIXTURES

Amina Khan -- CUS-1001

ACTIVE -- the primary demo customer for the interaction timeline.

Ravi Singh -- CUS-1002

PROSPECT to ACTIVE -- the onboarding and status-change journey.

CUS-9999

Not-found and negative-path testing -- never a real customer.

lab-request-001

The correlation ID threaded through every API call, event, and log.

KEY TAKEAWAY Fixture IDs CUS-1001, CUS-1002, CUS-9999, and lab-request-001 must appear consistently in every planning doc and every later lab.

Capstone Project Overview

1. Project Goal

- Solve a real-world problem
- Deliver a valuable, working solution
- Demonstrate enterprise-ready capabilities

2. Team Collaboration

- Work in cross-functional teams
- Define roles and responsibilities
- Communicate and collaborate effectively

3. Requirements & Planning

- Gather and analyze requirements
- Define scope and success criteria
- Create project plan and milestones

4. Solution Design

- Design system architecture
- Define components and integrations
- Plan data models and workflows

Build • Integrate • Deliver

Apply all the skills and knowledge gained throughout the program to build a complete, enterprise-ready solution.

5. Development & Integration

- Build and integrate application components
- Implement APIs and services
- Follow coding standards and best practices

6. Testing & Quality

- Perform unit, integration, and system testing
- Validate functionality and security
- Ensure performance and reliability

7. Deployment

- Automate CI/CD pipeline
- Deploy to target environments
- Validate deployment and monitoring

8. Presentation & Handover

- Present solution and demo
- Document architecture and decisions
- Share learnings and future enhancements

Success Criteria

- Meets all requirements
- High quality and secure
- Performance optimized
- Teamwork excellence
- Production ready

UNDERSTANDING BUSINESS REQUIREMENTS

Without named users, journeys, exclusions, and success measures, 'done' is contested in the Lab 52 defense panel.

WHAT TO CLARIFY

Primary users: service agent, manager, platform operator.

Journeys: search Amina/Ravi, view profile/timeline, record interaction.

In-scope capabilities vs. explicit exclusions (billing, real PII import).

Success measures tied to the Lab 52 demo and NFR thresholds.

OPEN QUESTIONS DISCIPLINE

- Every open question gets an owner and a due date.
- Vague 'build a CRM' gets rewritten with users and measurable success.
- Real customer names get replaced with synthetic fixtures immediately.
- A peer must be able to paraphrase the outcome statement in one pass.

context.md outcome statement (excerpt)

```
Users: service agent, manager, operator.
Journeys: search -> profile -> record interaction for CUS-1001.
Excluded: billing, real PII import, mobile app.
```

KEY TAKEAWAY A one-page outcome statement a peer can paraphrase, with exclusions explicit and every open question owned.

Understanding Business Requirements

The foundation of every successful solution.

1. Identify Stakeholders

Identify all stakeholders involved and understand their goals and expectations.

2. Elicit Requirements

Gather requirements through interviews, workshops, surveys, and document reviews.

3. Analyze & Clarify

Analyze, prioritize, and clarify requirements to ensure they are complete, clear, and consistent.

4. Document Requirements

Document requirements in a structured format for shared understanding and reference.

5. Validate Requirements

Review with stakeholders to ensure accuracy, feasibility, and alignment with business goals.

6. Manage Changes

Continuously track and manage changes to keep requirements relevant and aligned.

Why It Matters

- Aligns solutions with business goals
- Reduces risks and misunderstandings
- Improves planning and estimation
- Enhances stakeholder satisfaction
- Increases project success rate

FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

Functional requirements describe what the system does; non-functional requirements describe how well it must do it -- both need concrete acceptance criteria.

FUNCTIONAL REQUIREMENTS

Search and open a customer profile (Amina, Ravi).

Record a customer interaction (story CAP-12).

Change customer status (PROSPECT to ACTIVE).

View a customer's interaction timeline.

NON-FUNCTIONAL REQUIREMENTS (NFRs)

- Latency: p95 create-interaction API <= 500 ms in the lab.
- Availability: API readiness within 3 minutes after deploy.
- Security: unauthenticated /api/** returns 401; wrong role returns 403.
- Accessibility: keyboard-complete interaction form with associated labels.

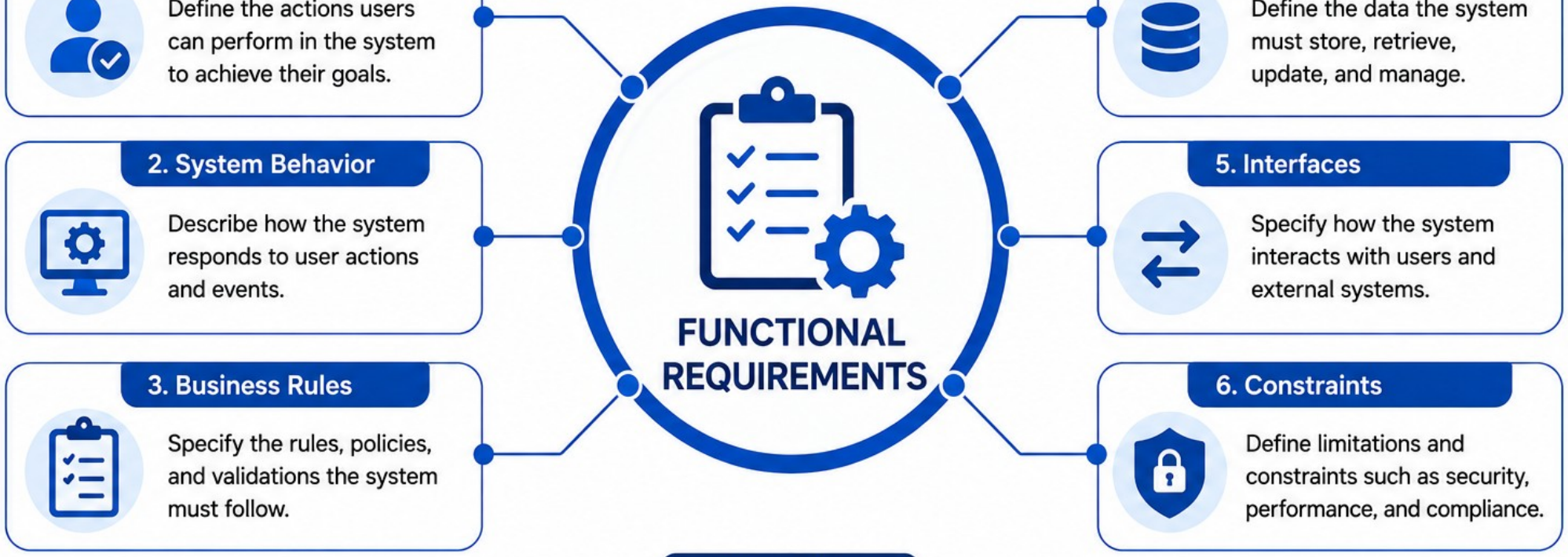
Banned vs. required language

```
Banned:   "the system should be fast and secure."
Required: "p95 latency <= 500 ms, measured via Micrometer, in the lab environment."
```

KEY TAKEAWAY 'Fast,' 'secure,' and 'scalable' without a threshold, method, and environment cannot be tested or defended.

Functional Requirements

Define **what** the system should do.



Examples

- User Registration and Login
- Add Items to Cart
- Generate Invoices
- Search and Filter Records
- Send Email Notifications
- Generate Reports

Non-Functional Requirements

Define **how** the system should perform.

1. Performance

The system should respond quickly and handle expected load efficiently.

2. Security

The system must protect data and ensure confidentiality, integrity, and availability.

3. Scalability

The system should handle growth in users, data, and transactions seamlessly.

4. Reliability

The system should operate consistently with minimal failures and downtime.

5. Availability

The system should be available and accessible when needed by users.

6. Usability

The system should be easy to use and provide a good user experience.

7. Maintainability

The system should be easy to maintain, update, and support.

8. Compliance

The system must comply with relevant laws, regulations, and standards.

Why It Matters

- Ensures quality and user satisfaction
- Improves performance and efficiency
- Protects data and systems
- Supports business growth
- Reduces maintenance and costs
- Ensures compliance and risk management

STAKEHOLDER EXPECTATIONS

Service agents, managers, and platform operators need different things from the same Northstar CRM release.

STAKEHOLDER NEEDS

Service agents: fast search, reliable interaction recording.

Managers: visibility into backlog priority and demo readiness.

Platform operators: safe deploy, rollback, and observability.

Instructors (Lab 52 panel): reproducible evidence, not claims.

MANAGING EXPECTATIONS

- State exclusions explicitly -- do not let scope creep in silently.
- Tie every promise to a backlog item or an ADR, never a verbal aside.
- Escalate blockers early, using the same rules learned in Module 47.
- Never present an unverified claim as fact to any stakeholder.

Traceability chain

```
Business outcome -> architecture choice -> backlog story -> test
-> deployment -> operations.
```

KEY TAKEAWAY Every stakeholder promise must be traceable to a backlog item or an ADR -- disconnected demos do not satisfy any of them.

Stakeholder Expectations

1. Business Sponsors

- Deliver business value and ROI
- Meet strategic objectives
- On-time and within budget
- Clear progress and outcomes

2. End Users

- Easy to use and reliable system
- Meets day-to-day needs
- Intuitive experience
- Responsive support

3. Project Managers

- Clear scope and requirements
- Realistic timelines and resources
- Risk mitigation and issue resolution
- Timely updates and collaboration

4. Development Team

- Well-defined requirements
- Technical clarity and feasibility
- Access to resources and tools
- Support and timely feedback

5. Operations / IT

- Stable and maintainable solution
- Easy deployment and monitoring
- High availability and performance
- Documentation and knowledge transfer

6. Quality Assurance

- Testable and measurable requirements
- Defect-free deliverables
- Quality standards and compliance
- Collaboration and environment support

7. Compliance / Legal

- Regulatory and policy compliance
- Data privacy and security
- Audit readiness
- Risk management

8. External Partners / Vendors

- Clear expectations and SLAs
- Timely communication
- Fair processes and evaluations
- Long-term partnership

ENTERPRISE ARCHITECTURE FUNDAMENTALS

C4 model views separate concerns so each audience sees the right level of detail.

Level	What It Shows
Context	People and external systems -- protocols and trust boundaries only.
Container	Deployable units -- React, Spring Boot, PostgreSQL, Kafka, the IdP.
Component	Modules within a container -- controllers, services, repositories.
Code	Classes and interfaces -- explicitly out of scope for Lab 48 diagrams.

WHY LEVELS MATTER

Context = No Internals

Keep class names and Kafka topic internals out of the context view.

Container = Deployment Units

Show synchronous REST+JWT and asynchronous event flows, both labeled.

Component = Later

Component and code views can wait until Lab 49 implementation begins.

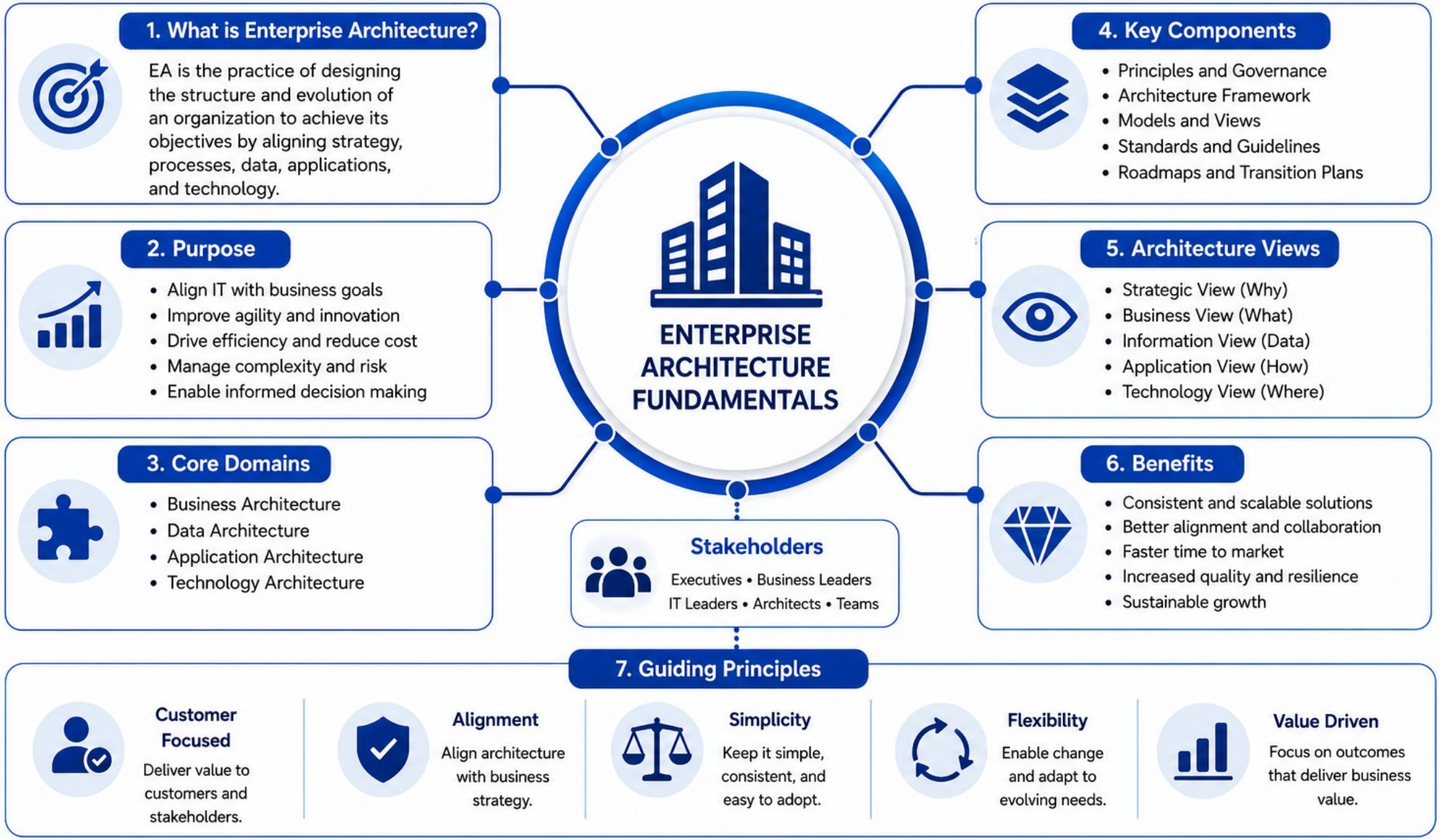
Trust Boundaries Everywhere

Mark where JWT auth, the database, and Kafka cross a trust zone.

KEY TAKEAWAY Context diagrams without trust boundaries hide IdP and data-exfiltration risks reviewers will probe in Lab 52.

Enterprise Architecture Fundamentals

A blueprint for aligning business strategy with technology.



IDENTIFYING SYSTEM COMPONENTS

Container placement forces the sync-versus-async decisions that ADRs and NFRs must later quantify.

CORE CONTAINERS

React CRM UI -- the agent-facing journey.

Spring Boot API -- REST + JWT resource server.

PostgreSQL -- system of record via Spring Data JPA.

Kafka + notification/worker consumer -- async side effects.

SUPPORTING SYSTEMS

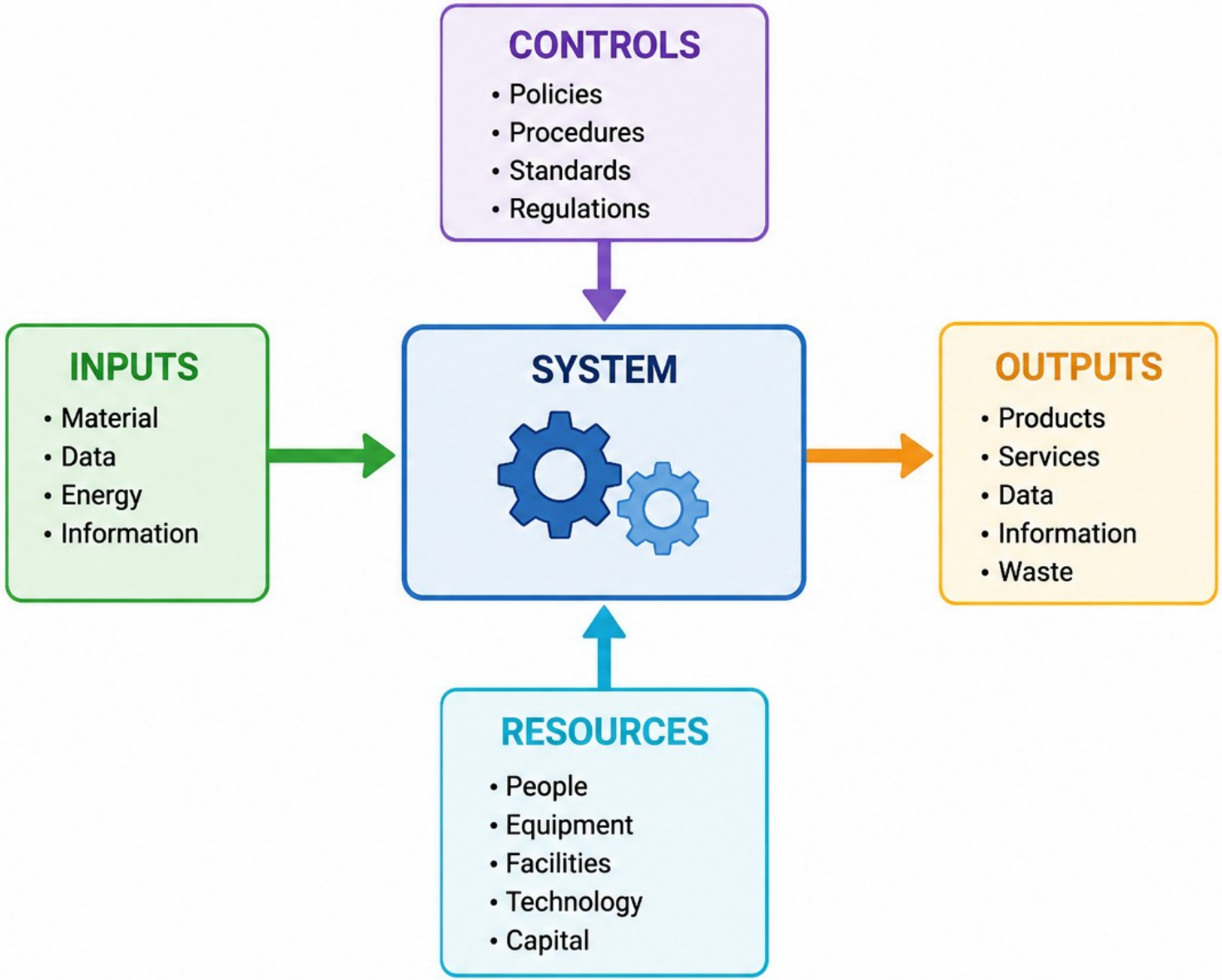
- Identity Provider -- OIDC/JWT issuance and validation.
- Logs / metrics / traces -- the observability container.
- Admin/deployment boundary -- who may apply k3s manifests.
- An orphan Kafka topic needs a documented consumer or a risk note.

container.md sync/async labels

```
UI --REST+JWT--> API --JPA/JDBC--> PostgreSQL
API --Customer events--> Kafka --> Notification Consumer
```

KEY TAKEAWAY Missing PostgreSQL or an orphan Kafka topic with no consumer are named Lab 48 failure modes -- add the container or document the risk.

Identifying System Components



TRY IT YOURSELF -- EXERCISE 1: SKETCH CONTEXT DIAGRAM

Reinforces: identifying users, external systems, and trust boundaries for Northstar CRM before any container-level detail is added.

STEPS (notes/lab48-context-sketch.md)

Step	What To Do
1 -- Actors	List service agents, admins, and external IdP/email/Kafka dependencies.
2 -- Check the Reference	Confirm docs/architecture/context.md and container.md are the target files.
3 -- Trust Boundaries	Mark where JWT auth, the database, and Kafka cross a trust zone.
4 -- Fixtures	Note CUS-1001/CUS-1002 as demo data, not external systems.

Context sketch skeleton

```
Actors: Service Agent, Manager, Platform Operator, Identity Provider.
Trust boundary: browser vs. API vs. IdP vs. Kafka consumers.
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 1 before continuing: exercises/exercise-01-context-sketch.md (workspace: examples/module-48-exercises).

FRONTEND / BACKEND / MESSAGING / DATABASE DESIGN

Four containers, four responsibilities -- each with its own contract to the others.

Frontend

React CRM UI -- agent search, profile, timeline, and interaction form over REST + JWT.

Backend

Spring Boot API -- layered controller/service/repository, Problem Details errors.

Messaging

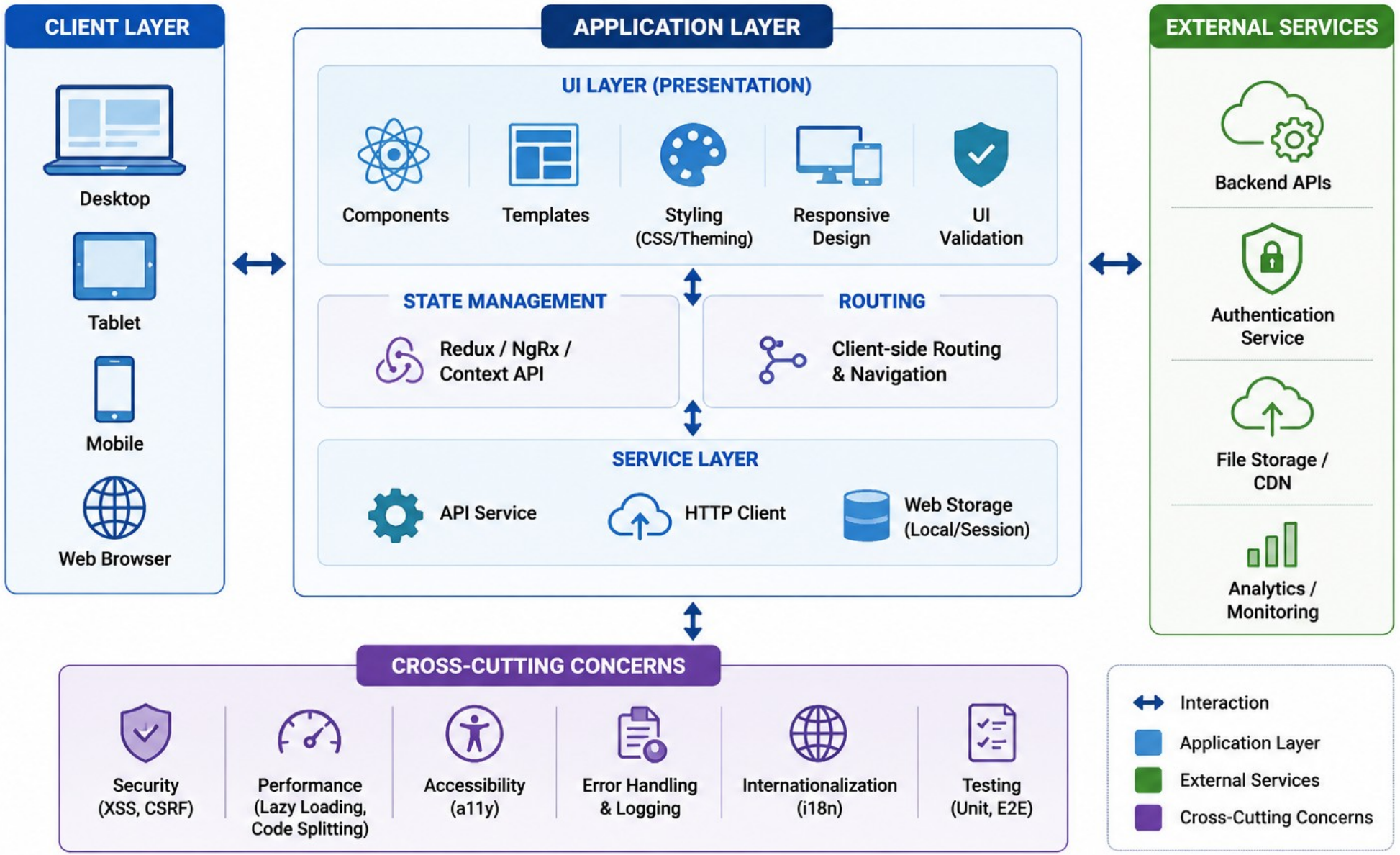
Kafka topic crm.customer.interactions.v1 -- CustomerInteractionRecordedV1 events.

Database

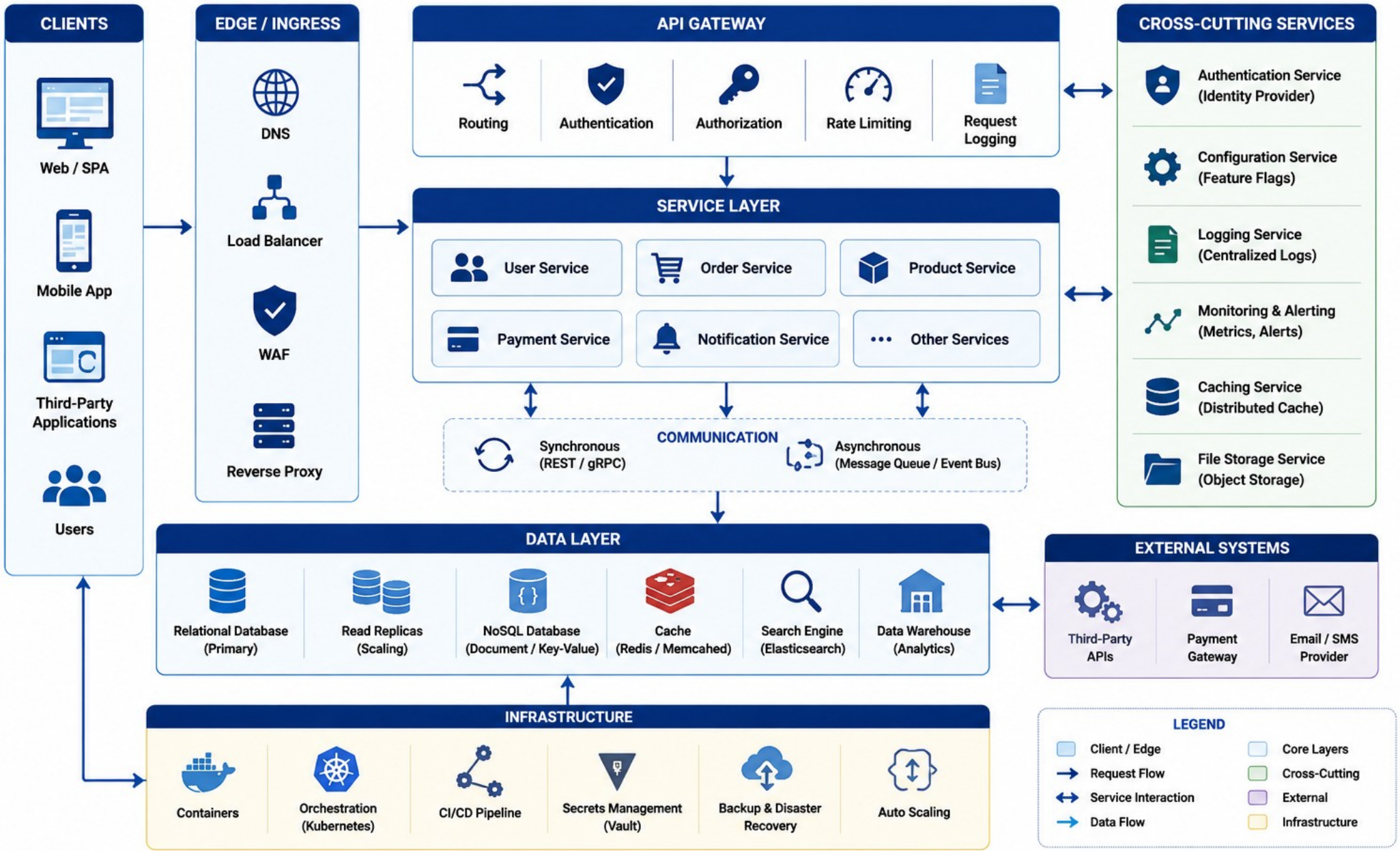
PostgreSQL system of record via Spring Data JPA -- migrations, not 'Hibernate will figure it out'.

KEY TAKEAWAY Each layer gets its own ADR-worthy decisions -- and Lab 49 implements exactly the backend and messaging halves of this picture.

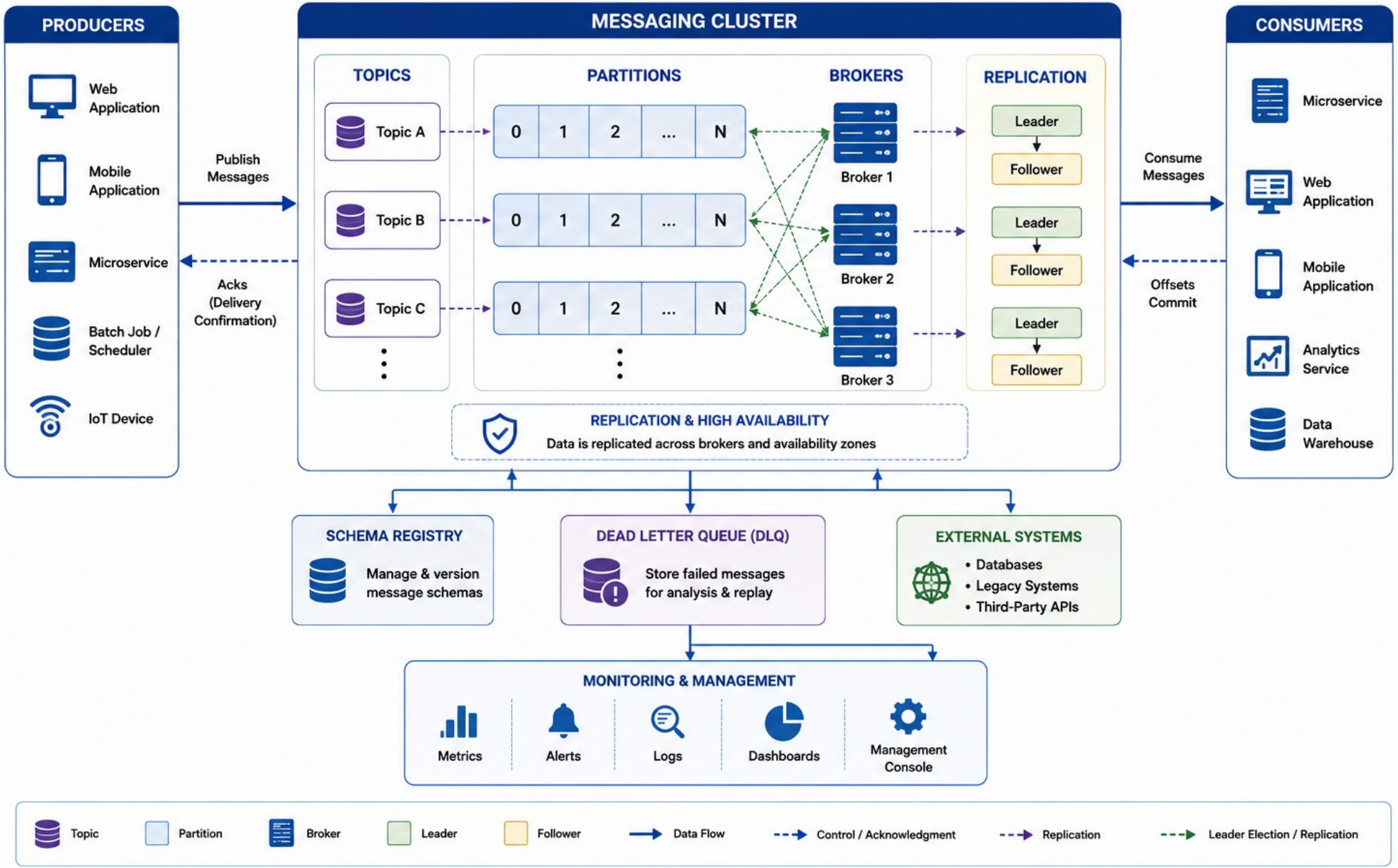
Frontend Architecture Design



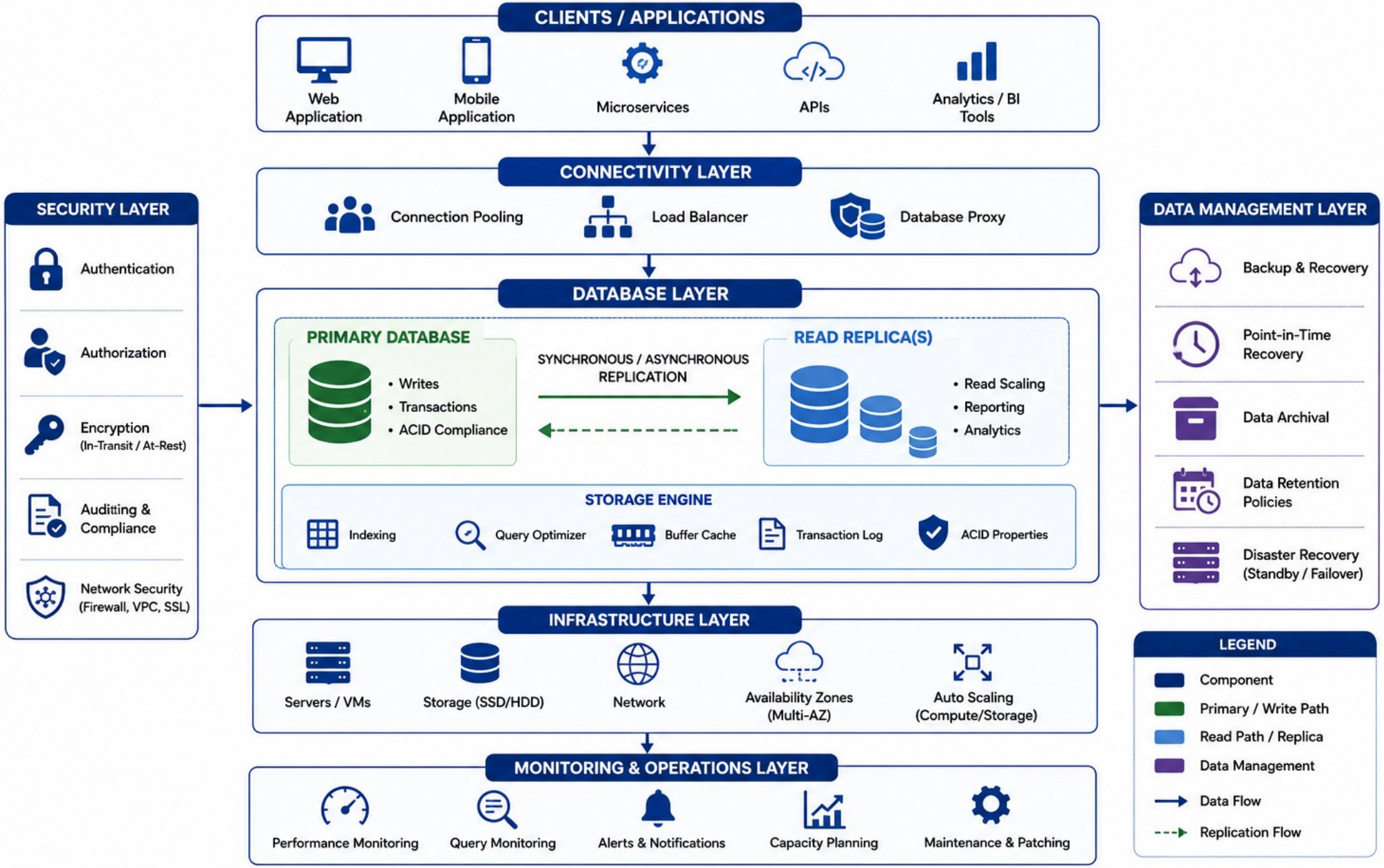
Backend Architecture Design



Messaging Architecture Design



Database Architecture Design



SELECTING THE TECHNOLOGY STACK

The Northstar CRM stack is fixed for Week 6 -- naming it precisely now avoids drift across Labs 49-51.

APPLICATION STACK

React (Vite, Node 22) for the agent-facing UI.

Spring Boot (JWT/RBAC) for the API layer.

PostgreSQL + Spring Data JPA for persistence.

Kafka for customer interaction events.

DELIVERY STACK

- Docker for immutable images.
- k3s (training cluster) for deployment; OpenShift as an enterprise comparison.
- GitHub Actions for CI/CD; Bitbucket Pipelines as an enterprise comparison.
- JUnit / Mockito for automated verification.

Stack-to-lab map

```
Lab 49: Spring Boot + Kafka.   Lab 50: React + PostgreSQL/JPA.
Lab 51: Docker + GitHub Actions + k3s.   Lab 52: defense + retrospective.
```

KEY TAKEAWAY Naming the stack once here prevents Lab 49-51 from re-arguing technology choices mid-week.

Selecting the Technology Stack

INPUTS

- Business Goals
- Functional Requirements
- Non-Functional Requirements
- Stakeholder Expectations
- Budget & Timeline

TECHNOLOGY STACK LAYERS

Frontend	React	Angular	Vue.js	Next.js
Backend	Spring Boot	Node.js	Django	.NET
Database	PostgreSQL	MySQL	MongoDB	Redis
Cloud & Infrastructure	aws	Microsoft Azure	Google Cloud	Docker
DevOps & Tools	GitHub	Jenkins	Kubernetes	Terraform
Security	Keycloak	OAuth 2.0	HashiCorp Vault	OWASP

EVALUATION CRITERIA

- Performance
- Security
- Scalability
- Cost
- Community & Support
- Integration
- Time to Market
- Maintainability

GUIDING PRINCIPLES

- Align with Business Goals
- Prioritize Simplicity
- Ensure Scalability
- Build for Security
- Enable Flexibility
- Optimize Cost
- Use Proven & Reliable Tech

DEFINING SYSTEM BOUNDARIES

A clear boundary tells Labs 49-52 what is this week's problem and what is explicitly deferred.

IN SCOPE

Customer search, profile, and interaction timeline.

Recording an interaction (story CAP-12) end to end.

Status change: PROSPECT to ACTIVE.

A secured, deployed release with rollback evidence.

OUT OF SCOPE

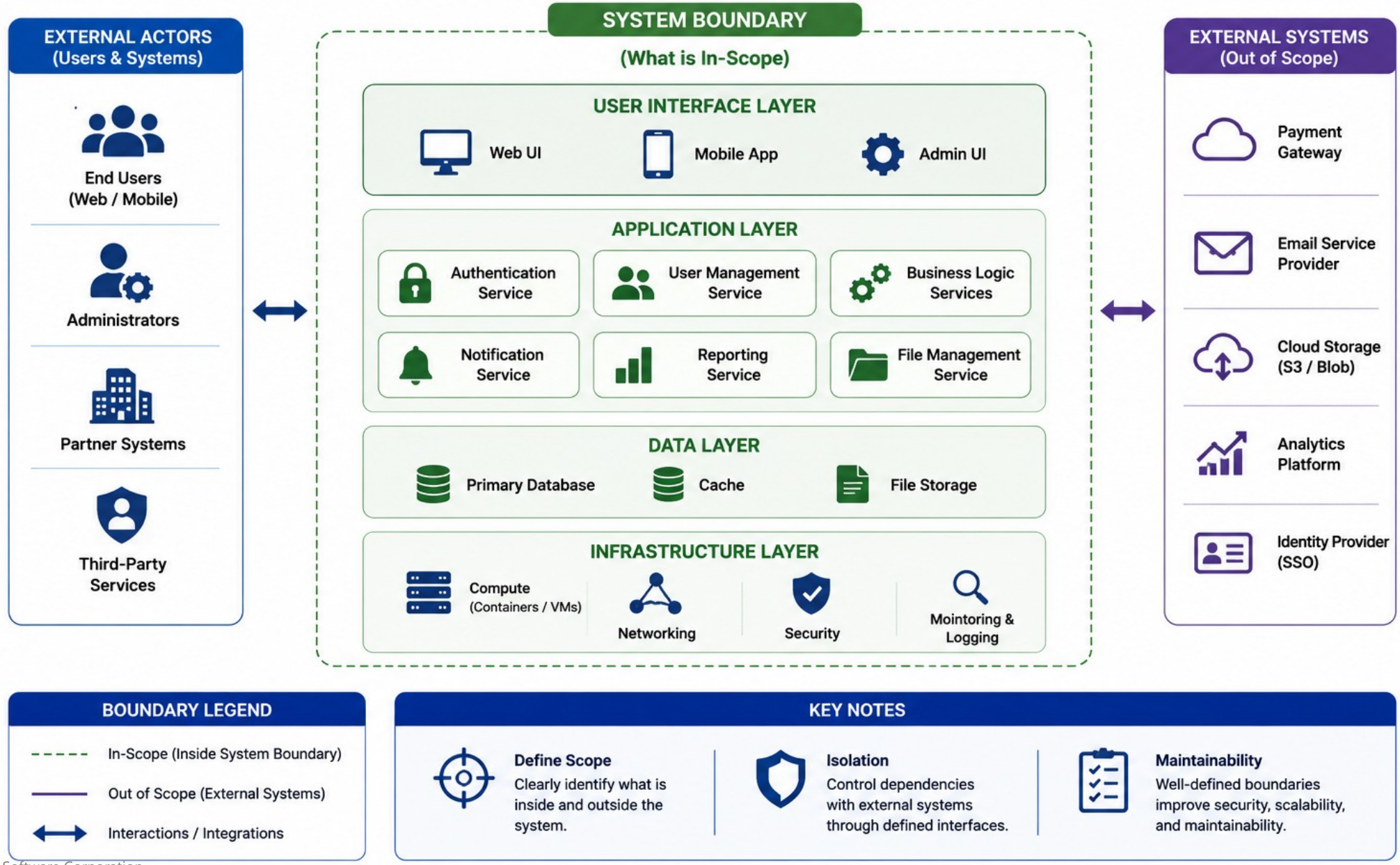
- Billing and payments.
- Real customer PII import from production systems.
- A native mobile app.
- Multi-tenant white-labeling.

Explicit deferral note

```
"Billing is explicitly out of scope for Week 6 -- deferred, not silently dropped."
```

KEY TAKEAWAY If a story is deferred, mark it Explicitly Deferred with an owner and date in the risk register -- never silently drop it.

Defining System Boundaries



BOUNDARY LEGEND

- In-Scope (Inside System Boundary)
- Out of Scope (External Systems)
- Interactions / Integrations

KEY NOTES

Define Scope
Clearly identify what is inside and outside the system.

Isolation
Control dependencies with external systems through defined interfaces.

Maintainability
Well-defined boundaries improve security, scalability, and maintainability.

IDENTIFYING INTEGRATION POINTS

Every integration point is a place contract drift or a trust-boundary failure can hide.

INTEGRATION POINTS

UI <-> API over REST + JWT.

API <-> PostgreSQL over JPA/JDBC.

API <-> Kafka for customer events.

Pipeline <-> registry <-> cluster for deploy.

CONTRACT DISCIPLINE

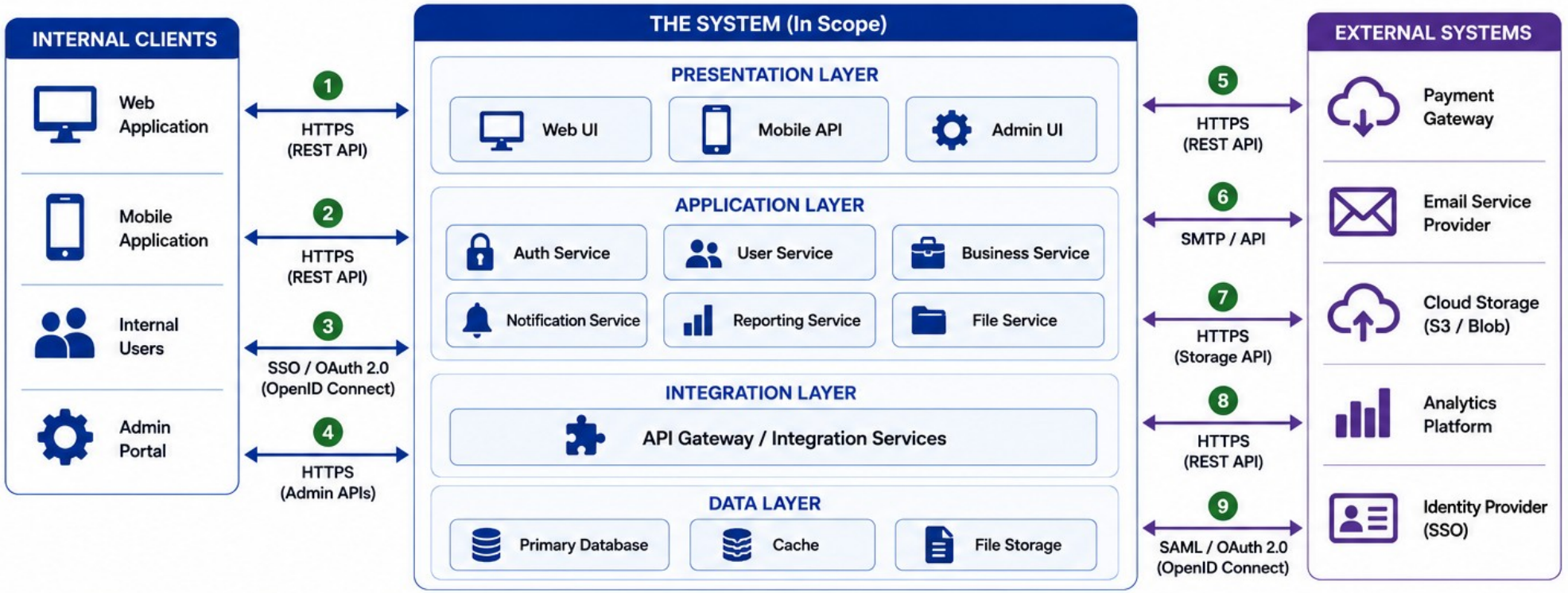
- Draft POST /api/customers/{id}/interactions with Problem Details errors.
- Draft CustomerInteractionRecordedV1 fields before Lab 49 codes them.
- Additive fields are fine; breaking changes require a version bump.
- Every request carries header X-Correlation-ID: lab-request-001.

Event sketch (fields only)

```
eventId, eventType, eventVersion, occurredAt,
correlationId, actor, customerId, interactionId, channel
```

KEY TAKEAWAY Unversioned endpoints and events create Lab 50 contract breakage and Lab 52 panel failure -- version from day one.

Identifying Integration Points



INTEGRATION POINTS SUMMARY

ID	Integration Point	Type	Protocol / Technology	Direction	Purpose
1	Web Application ↔ System	API	HTTPS (REST API)	↔	Access system features and data
2	Mobile Application ↔ System	API	HTTPS (REST API)	↔	Access system features and data
3	Internal Users ↔ System (SSO)	Security	SSO / OAuth 2.0 (OIDC)	↔	Authenticate and authorize users
4	Admin Portal ↔ System	API	HTTPS (Admin APIs)	↔	System administration
5	System ↔ Payment Gateway	API	HTTPS (REST API)	↔	Process payments and refunds
6	System ↔ Email Provider	Service	SMTP / API	→	Send emails and notifications
7	System ↔ Cloud Storage	API	HTTPS (Storage API)	↔	Store and retrieve files
8	System ↔ Analytics Platform	API	HTTPS (REST API)	→	Send events and metrics
9	System ↔ Identity Provider	Security	SAML / OAuth 2.0 (OIDC)	↔	SSO authentication and user info

LEGEND

- Internal Components (In Scope)
- External Systems (Out of Scope)
- N — Integration Point ID
- ↔ Bi-directional Integration
- → Outbound Integration
- ↔ External Integration
- Integration Layer (Orchestration)

SECURITY ARCHITECTURE PLANNING

Security decisions made now save Lab 51 from a redesign under deadline pressure.

Concern	Lab 48 Decision
Authentication	OIDC/JWT resource-server pattern, decided via ADR.
Authorization	Role-based -- AGENT and MANAGER roles at minimum.
Secrets	Never in ADRs, screenshots, or committed docs.
Untrusted Input	Browser agents, JWT claims, and Kafka payloads are all untrusted.

SECURITY PLANNING HABITS

JWT/OIDC ADR Now

Decide the auth pattern in Lab 48 -- Lab 51 hardens, it does not redesign.

Role-Based, Not Person-Based

Plan AGENT/MANAGER roles, never per-person special cases.

Fictional Emails Only

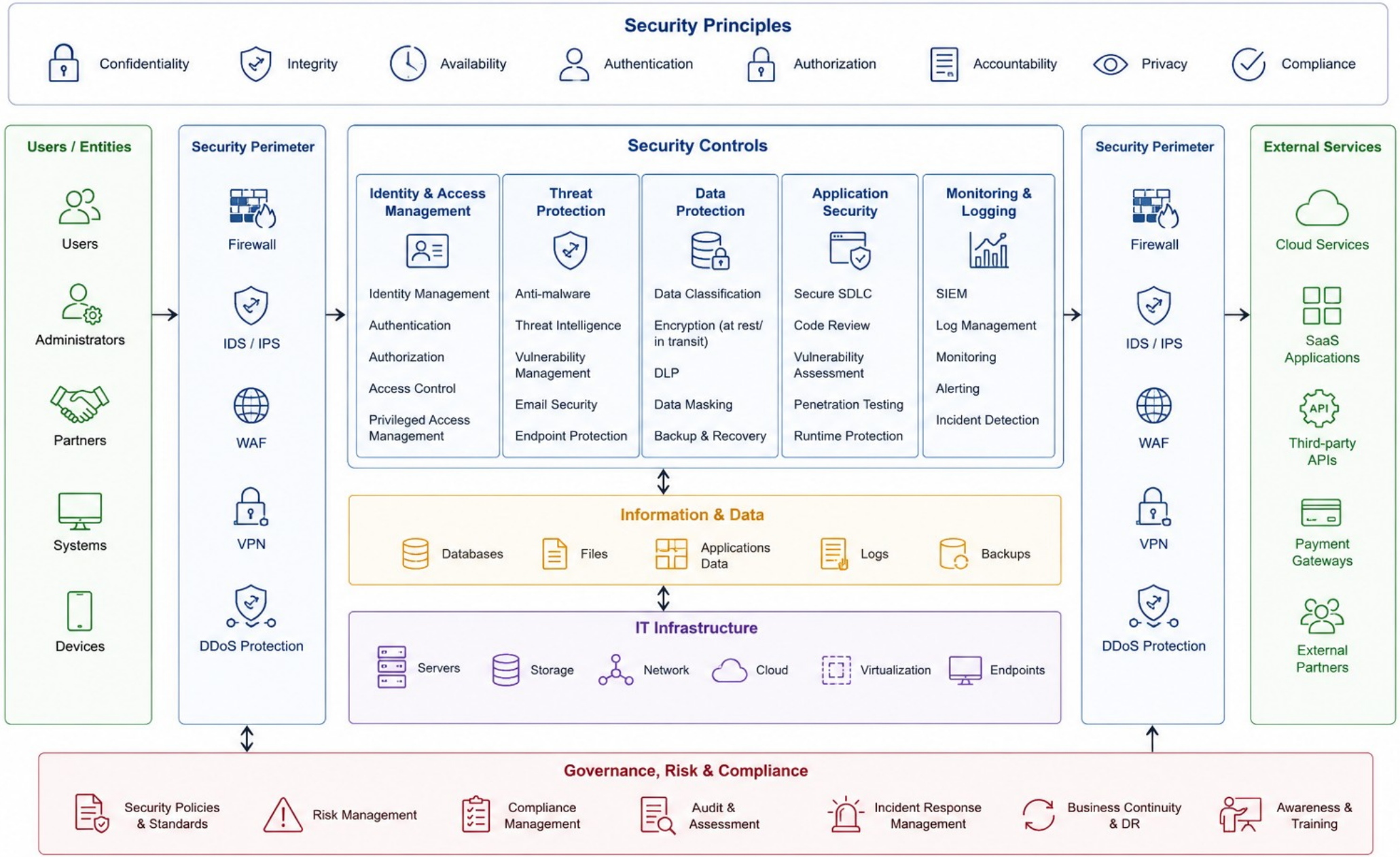
amina.khan@example.test, never a real email, in any planning doc.

No Secrets In Docs

Never paste real IdP secrets, kubeconfigs, or production URLs.

KEY TAKEAWAY Which inputs are untrusted, where authn/authz is enforced, and which values are sensitive -- answer all three before Lab 49 starts coding.

Security Architecture Planning



SCALABILITY / RELIABILITY CONSIDERATIONS

Scalability and reliability are NFRs too -- they need thresholds and a measurement method, not adjectives.

SCALABILITY

Partition Kafka events by customer ID for ordered processing.

Avoid N+1 queries when loading interaction timelines.

Index PostgreSQL for timeline-by-customer-and-time lookups.

Document which profile targets PostgreSQL vs. H2 in CI.

RELIABILITY

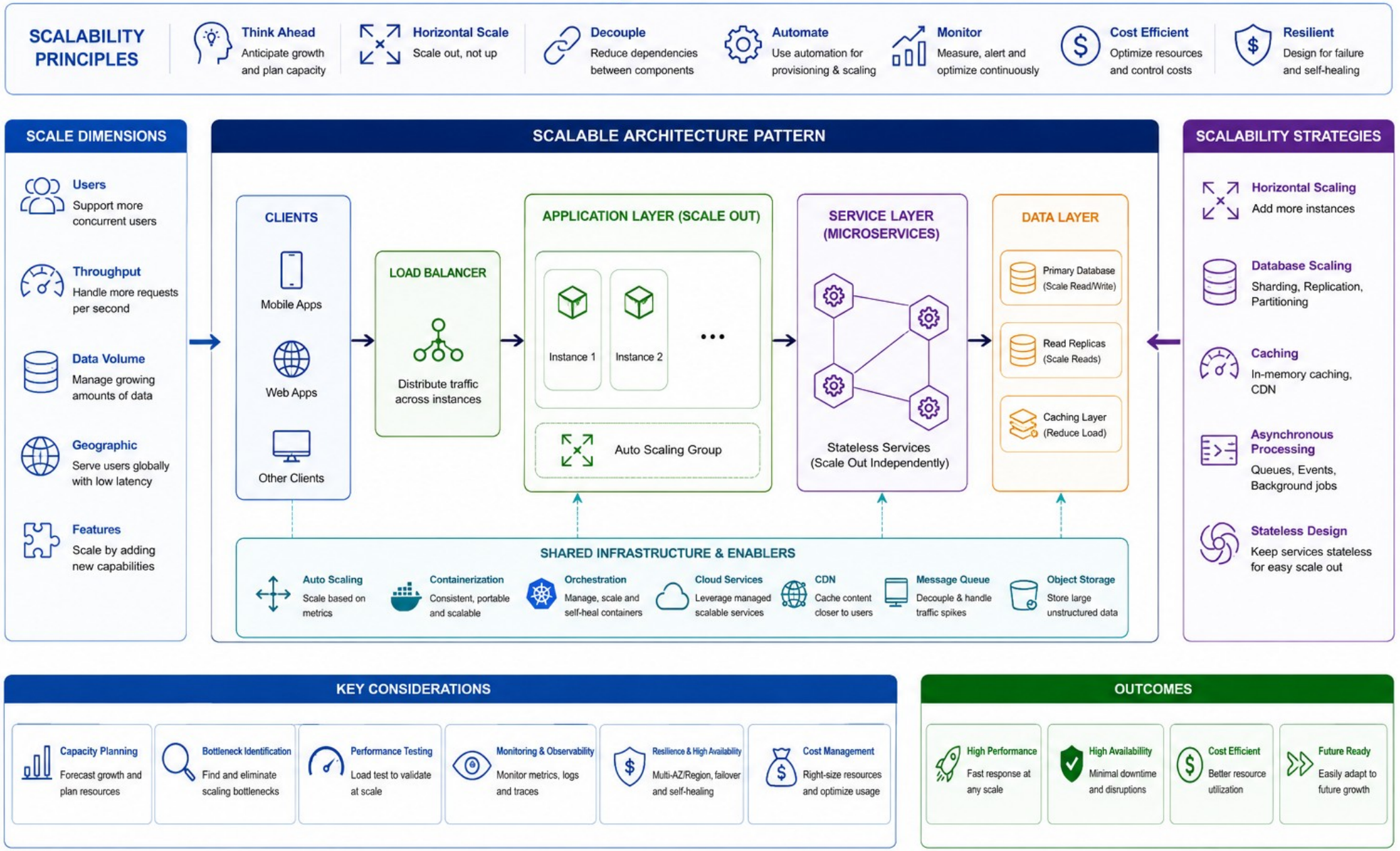
- Rollback the previous digest within 10 minutes (rehearsed in Lab 51).
- Consumers must be idempotent -- duplicate events are normal.
- Bounded retries plus a dead letter topic for poison messages.
- API readiness probe within 3 minutes after deploy.

NFR line (reliability)

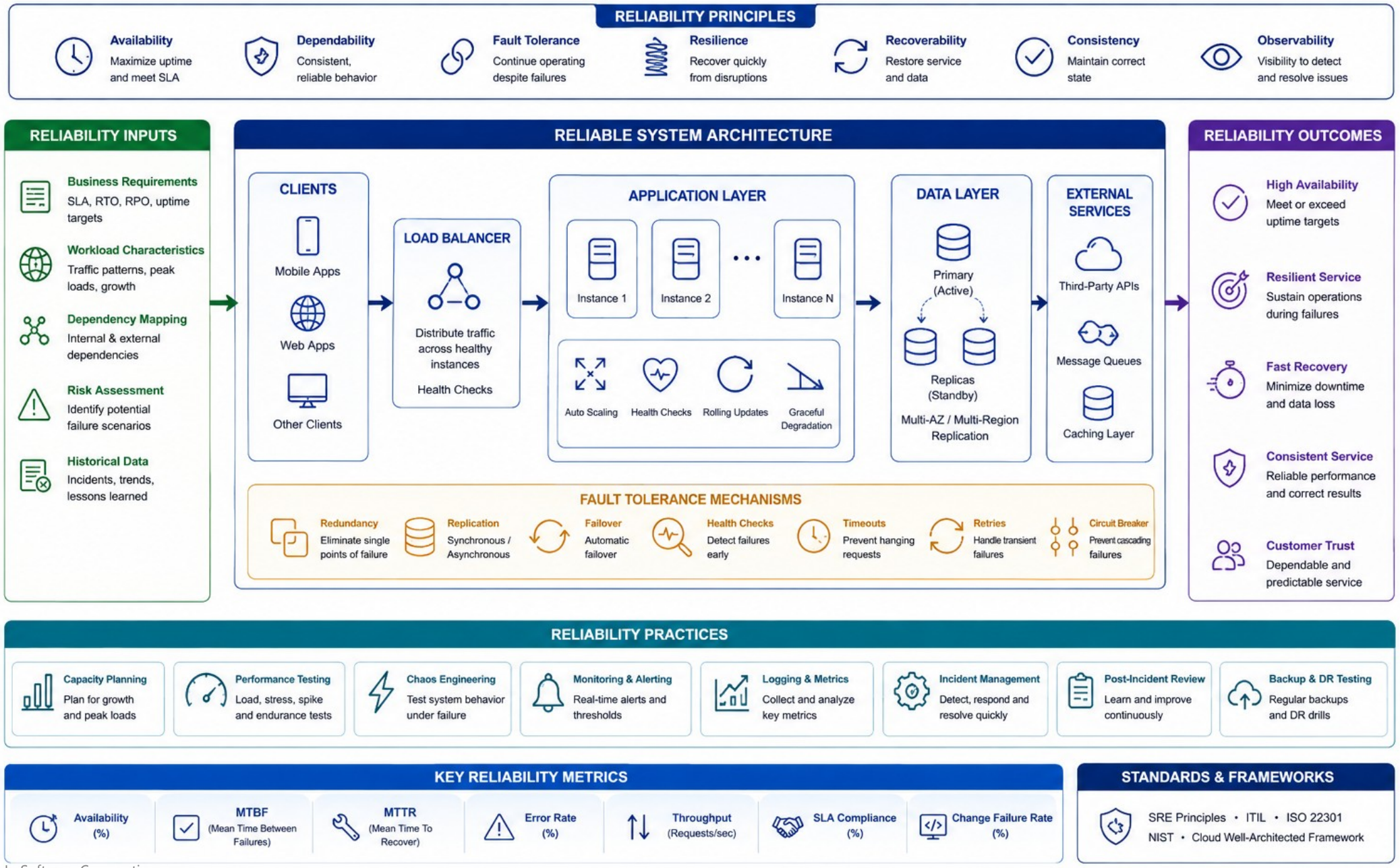
```
"Rollback previous digest <= 10 min, method: rehearse in Lab 51, environment: k3s training namespace."
```

KEY TAKEAWAY Every scalability or reliability claim needs a number, a measurement method, and an environment -- or it is not an NFR yet.

Scalability Considerations



Reliability Considerations



TRY IT YOURSELF -- EXERCISE 2: DRAFT MEASURABLE NFRS

Reinforces: replacing vague quality words with measurable NFR placeholders across security, traceability, recoverability, performance, and operability.

STEPS (notes/lab48-nfr-placeholders.md)

Step	What To Do
1 -- Categories	Security, traceability, recoverability, performance, operability -- one NFR each.
2 -- Check the Reference	Each NFR needs metric + target + how you will prove it in Labs 49-52.
3 -- Example Rewrite	Rewrite 'system should be fast' into a measurable statement with a placeholder number.
4 -- Scope	Placeholders only -- the full NFR doc is completed in Lab 48.

Rewrite pattern

```
Vague: "the system should be fast."
Measurable: "p95 latency <= ___ ms, measured via ___, in ___."
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 2 before continuing: exercises/exercise-02-nfr-placeholders.md (workspace: examples/module-48-exercises).

AGILE PROJECT PLANNING

The capstone runs on the same Agile rhythm as every prior module, scaled up to a five-lab delivery week.

Role	Responsibility	Communicates With
Product Owner	Owns backlog priority for CAP-12 and friends.	Stakeholders and the team.
Scrum Master	Facilitates process, removes blockers.	Team and leadership.
Development Team	Delivers Labs 49-51's vertical slices.	Product Owner and QA.
Release Commander	Owns the Lab 51 deploy and rollback decision.	Ops, engineers, stakeholders.

WEEK 6 SPRINT CYCLE

1

1. Plan (Lab 48)

Architecture, NFRs, backlog, ADRs, risk register.

2

2. Build (Lab 49-50)

Backend and Kafka, then React and PostgreSQL.

3

3. Release (Lab 51)

Security, CI/CD, containers, k3s deploy.

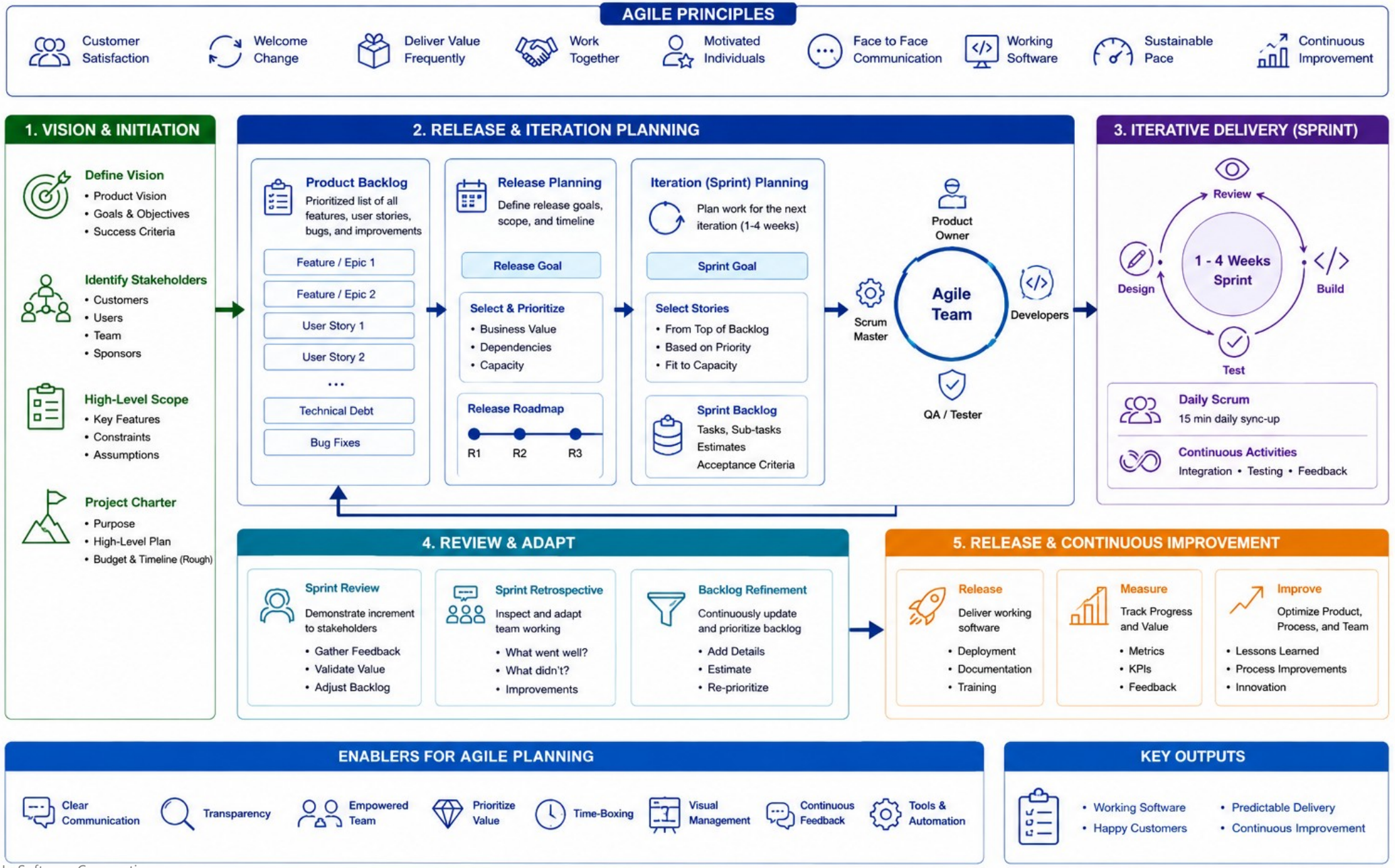
4

4. Defend (Lab 52)

Live demo, Q&A, retrospective, rubric score.

KEY TAKEAWAY Kanban visualizes flow with a work-in-progress limit; this capstone timeboxes flow into five labs -- both exist to keep delivery predictable.

Agile Project Planning



USER STORIES AND BACKLOG CREATION

Horizontal 'layers first' backlogs strand Week 6 without a demoable vertical slice -- every story must cross API, UI, data, and events lightly.

VERTICAL STORY RULE

Order by value, risk, dependency, and learning -- not by layer.

Every story needs numbered acceptance criteria, not just a task title.

Enabling tech stories must cite which outcome they unlock.

Amina/Ravi fixtures appear in acceptance criteria, not random data.

CAP-12 WORKED EXAMPLE

- As a service agent, I want to record an interaction for CUS-1001.
- AC1: valid input returns 201 + a resource id; correlation preserved.
- AC2: the timeline shows the interaction within two seconds of refresh.
- AC4: invalid notes return field-level errors and are not persisted.

CAP-12 -- Record a customer interaction

```
As a service agent, I want to record an interaction for CUS-1001
(Amina Khan) so the next agent understands customer history.
```

KEY TAKEAWAY CAP-12 is the backbone story Lab 49 implements end to end -- get its acceptance criteria right here, or Lab 49 improvises.

User Stories and Backlog Creation

1. WHAT IS A USER STORY?

A user story is a short, simple description of a feature from the perspective of the user.

User Story Template

As a <type of user>,
I want <an action>,
so that <a benefit>.

Example

As a registered user,
I want to reset my password
so that I can regain access
to my account.

Characteristics

- **Independent** — Can stand on its own
- **Negotiable** — Open to discussion and change
- **Valuable** — Provides value to the user/business
- **Estimable** — Can be estimated in size
- **Small** — Small enough to complete in a sprint
- **Testable** — Can be verified

2. FROM IDEA TO USER STORIES

1. **Gather Requirements**
   - Interviews
   - Workshops
   - Surveys
   - Existing Systems
   - Feedback
2. **Identify Epics** — Group related requirements into epics (large features).
3. **Write User Stories** — Break epics into user stories using the user story template.
4. **Define Acceptance Criteria**
   - Clear
   - Testable
   - Measurable
   - Agreed by all
5. **Estimate (Sizing)** — Estimate effort/size using planning poker, T-shirt sizing, or story points.

EXAMPLE: EPIC BREAKDOWN

Epic: User Account Management

Story 1	Story 2	Story 3	Story 4	Story 5
User Registration	User Login	Reset Password	Update Profile	Deactivate Account
Acceptance Criteria	Acceptance Criteria	Acceptance Criteria	Acceptance Criteria	Acceptance Criteria
• ...	• ...	• ...	• ...	• ...
• ...	• ...	• ...	• ...	• ...
• ...	• ...	• ...	• ...	• ...

3. BACKLOG CREATION

Product Backlog

A prioritized list of user stories and other items needed to deliver value.

Backlog Items

US-001 — As a registered user, I want to reset my password so that I can regain access to my account. (5)

US-002 — As a new user, I want to register an account so that I can use the system. (8)

US-003 — As a user, I want to update my profile so that my information is accurate. (3)

• • •

Higher Priority ↕ Lower Priority

4. BACKLOG REFINEMENT (ONGOING)

1. **Review & Clarify** — Ensure understanding of the story, intent, and value.
2. **Add Details** — Elaborate acceptance criteria, notes, and edge cases.
3. **Re-estimate** — Refine size as more information becomes available.
4. **Re-prioritize** — Adjust priority based on value, risk, and dependencies.
5. **Ready for Sprint** — Stories at the top are ready for sprint planning.

5. BENEFITS

- Clear understanding of user needs
- Better estimation and planning
- Prioritized delivery of value
- Improved collaboration
- Higher customer satisfaction

TRY IT YOURSELF -- EXERCISE 3: SKETCH VERTICAL STORIES

Reinforces: writing three vertical stories that cross API/UI/data/events lightly, with acceptance criteria and Must/Should priority.

STEPS (notes/lab48-backlog-slice.md)

Step	What To Do
1 -- Stories	Create customer, record interaction, search + timeline -- using Amina/Ravi fixtures.
2 -- Check the Reference	Stories need acceptance criteria -- not tasks like 'make controller'.
3 -- Priority	Order Must/Should and note dependencies on Labs 49-51.
4 -- Scope	Three stories only -- the full backlog is completed in Lab 48.

Story skeleton

```
As a <role>, I want <capability> so <outcome>.
Acceptance criteria: 1) ...   2) ...   3) ...
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 3 before continuing: exercises/exercise-03-backlog-slice.md (workspace: examples/module-48-exercises).

TASK BREAKDOWN STRUCTURE

A vertical story still needs a concrete task list -- broken down without losing its vertical shape.

BREAKING DOWN CAP-12

DTOs + Bean Validation for the interaction request.

Flyway migration for the customer_interaction table.

Transactional InteractionService with correlation propagation.

REST endpoint + Kafka publisher + consumer + tests.

BREAKDOWN RULES

- Tasks stay inside one story -- do not silently expand scope.
- Each task should be completable in under a day.
- Tests are a task, not an afterthought.
- Documentation (backend-demo.md) is a task, not optional.

Task list excerpt (CAP-12)

```
1) DTOs   2) Migration   3) Service   4) Endpoint
5) Publisher   6) Consumer   7) Tests   8) backend-demo.md
```

KEY TAKEAWAY A good breakdown turns 'implement CAP-12' into eight completable tasks without losing the vertical, demoable shape.

Task Breakdown Structure

What is a TBS? A Task Breakdown Structure (TBS) is a hierarchical decomposition of the total project work into smaller, manageable tasks.

- Improves Planning
- Enhances Estimation
- Clarifies Ownership
- Increases Visibility

PROJECT
Online Shopping Platform

- **1. INITIATION**
  - 1.1 Define Project Objectives
  - 1.2 Identify Stakeholders
  - 1.3 Feasibility Analysis
  - 1.4 Project Charter
- **2. PLANNING**
  - 2.1 Requirements Gathering
  - 2.2 Scope Definition
  - 2.3 Create Project Plan
  - 2.4 Resource Planning
  - 2.5 Risk Assessment
  - 2.6 Schedule Planning
- **3. EXECUTION**
  - **3.1 DESIGN**
    - 3.1.1 System Architecture
    - 3.1.2 Database Design
    - 3.1.3 UI/UX Design
    - 3.1.4 Design Review
  - **3.2 DEVELOPMENT**
    - 3.2.1 Frontend Development
    - 3.2.2 Backend Development
    - 3.2.3 API Development
    - 3.2.4 Integration
  - **3.3 SUPPORTING TASKS**
    - 3.3.1 Setup Environment
    - 3.3.2 Code Review
    - 3.3.3 Documentation
    - 3.3.4 Team Meetings
- **4. TESTING**
  - 4.1 Test Planning
  - 4.2 Test Case Design
  - 4.3 Functional Testing
  - 4.4 Performance Testing
  - 4.5 Security Testing
  - 4.6 UAT
- **5. DEPLOYMENT**
  - 5.1 Deployment Planning
  - 5.2 Environment Setup
  - 5.3 Build & Release
  - 5.4 Deployment
  - 5.5 Release Verification
  - 5.6 Rollback Plan
- **6. MAINTENANCE**
  - 6.1 Monitor System
  - 6.2 Incident Management
  - 6.3 Bug Fixes
  - 6.4 Performance Optimization
  - 6.5 Enhancements
  - 6.6 Documentation Updates

TBS LEVELS EXAMPLE

LEVEL 1 Project (Total Project) → **LEVEL 2** Phases (Major Deliverables) → **LEVEL 3** Sub-Phases (Key Areas of Work) → **LEVEL 4** Tasks (Work Packages) → **LEVEL 5** Sub-Tasks (Detailed Activities)

KEY POINTS

- Break work down to manageable tasks
- Each task should have a clear owner
- Tasks should be measurable and estimable
- Decompose until the right level of detail is achieved

TEAM ROLES AND RESPONSIBILITIES

Named owners, not a diffuse 'the team,' keep Labs 49-52 from renegotiating scope mid-week.

Lab 49 Owner

CAP-12 API + Kafka for CUS-1001, with correlation lab-request-001.

Lab 50 Owner

React search/profile/timeline + PostgreSQL durability proof.

Lab 51 Owner

JWT deny-by-default, pipeline gates, immutable image, rollback.

Lab 52 Owner

Demo script, evidence index, Q&A cards, retro, self-assessment.

KEY TAKEAWAY Write this ownership checklist into docs/team-plan.md so Week 6 does not thrash over who owns what.

Team Roles and Responsibilities

Collaborate — Work together as one team

Deliver Value — Focus on customer value

Own Results — Take ownership and accountability

Communicate — Open, transparent communication

Adapt — Embrace change and improve

Respect — Value every contribution

ROLE	PRIMARY RESPONSIBILITIES	KEY ACCOUNTABILITIES	KEY SKILLS / COMPETENCIES	WORKS CLOSELY WITH
Product Owner	• Define product vision and strategy • Manage and prioritize the product backlog • Represent stakeholders and customer needs • Accept or reject work items	• Value delivery • Backlog ROI and prioritization • Clear requirements and acceptance • Stakeholder satisfaction	• Business analysis • Prioritization & decision making • Communication & negotiation • Domain knowledge	Stakeholders Scrum Master Development Team
Scrum Master	• Facilitate Agile ceremonies • Remove impediments for the team • Coach the team on Agile principles • Promote continuous improvement	• Team effectiveness • Process adherence • Impediment resolution • Continuous improvement	• Agile practices • Facilitation • Coaching & mentoring • Problem solving	Product Owner Development Team Organization
Development Team	• Design, build, test, and deliver increments • Collaborate to meet sprint goals • Participate in planning and estimation • Maintain code quality and documentation	• Commitments • Quality of deliverables • Velocity and predictability • Technical excellence	• Technical expertise • Collaboration • Problem solving • Adaptability	Product Owner Scrum Master QA / Testers
QA / Tester	• Create test strategy and test plans • Design and execute test cases • Identify defects and verify fixes • Ensure quality standards	• Defect detection • Test coverage • Quality of releases • Timely feedback	• Testing techniques • Automation tools • Attention to detail • Analytical thinking	Development Team Product Owner DevOps
DevOps Engineer	• Build and maintain CI/CD pipelines • Manage infrastructure and deployments • Monitor systems and ensure availability • Automate processes and tools	• Deployment frequency • System reliability • Build & release quality • Incident response time	• CI/CD & Automation • Cloud & Infrastructure • Scripting / IaC • Monitoring & Security	Development Team QA / Testers Operations
UI/UX Designer	• Understand user needs and behaviors • Create wireframes, prototypes, and designs • Ensure usability and accessibility • Collaborate on user-centered solutions	• Usability • Design quality • User satisfaction • Consistent UX	• UI/UX Design • Prototyping tools • User research • Communication	Product Owner Development Team QA / Testers
Business Analyst	• Elicit and document requirements • Analyze processes and data • Create user stories and use cases • Validate solution meets business needs	• Requirement accuracy • Traceability • Business alignment • Stakeholder communication	• Requirements analysis • Modeling & documentation • Communication • Domain knowledge	Product Owner Development Team Stakeholders

SPRINT PLANNING FOR CAPSTONE

Five labs, five days -- sprint planning here is really a critical-path plan across the whole capstone week.

CRITICAL PATH

Lab 48 architecture and backlog gate everything downstream.

Lab 49 backend/Kafka must land before Lab 50's UI can integrate.

Lab 51 security/deploy needs a working Lab 49-50 slice.

Lab 52 defense needs evidence from every prior lab.

PLANNING DISCIPLINE

- Time-box diagramming and ADR writing -- they often burn 90 minutes.
- Require one complete vertical story early, not five partial ones.
- Track milestones with named owners and due dates.
- Re-plan explicitly if a story is deferred -- never silently drop it.

Milestone line

```
"CAP-12 API+Kafka green by end of Lab 49 -- owner: <name>, due: Day 2."
```

KEY TAKEAWAY The critical path runs Plan -> Backend -> Frontend -> Release -> Defense -- a slip anywhere upstream delays the whole chain.

Sprint Planning for Capstone

PURPOSE

Define sprint goal, select capstone backlog items, estimate, and create a commitment plan.

OUTCOMES

- Clear Sprint Goal
- Selected & Prioritized Backlog Items
- Estimated Work
- Team Commitment
- Sprint Plan Ready

1. INPUTS

Capstone Product Vision
High-level objectives and success criteria

Product Backlog
Prioritized list of features, stories, tasks, and NFRs

Team Capacity
Team availability and skills for the sprint

Constraints & Dependencies
Technical, resource, and external dependencies

Definition of Done
Quality standards and completion criteria

2. SPRINT PLANNING ACTIVITIES

1. Set the Sprint Goal
- Define what the team aims to achieve in this sprint
- Align with capstone objectives

2. Review & Select Backlog Items
- Review top-priority items
- Discuss scope and clarify requirements

3. Estimate the Work
- Break down into tasks
- Estimate using story points or hours

4. Plan the Work
- Decide how the work will be done
- Identify tasks, owners, and dependencies

5. Confirm Commitment
- Validate capacity vs. commitment
- Finalize sprint backlog

3. SPRINT PLAN OUTPUTS

SPRINT GOAL
A short, clear statement of what the team will achieve in the sprint.

SPRINT BACKLOG

ID	Backlog Item / Task	Estimate	Owner
US-101	Implement User Authentication	8	Alex
US-102	Build Product Catalog API	13	Priya
US-103	Develop React Product List	8	Sam
US-104	Cart & Checkout Flow	13	Jordan
US-105	Write Unit & Integration Tests	8	Taylor
...	...	...	...

PLAN SUMMARY

KEY DETAILS

Sprint	Duration	Team Size	Total Points	Capacity
Sprint 1	2 Weeks	6	50	50

4. SPRINT PLANNING CHECKLIST

- [ ] Sprint goal is clear and aligned with capstone vision
- [ ] Backlog items are prioritized and well understood
- [ ] Stories are broken down into actionable tasks
- [ ] All items are estimated
- [ ] Dependencies and risks are identified
- [ ] Team has the skills and capacity to deliver
- [ ] Commitment is confirmed
- [ ] Sprint backlog is created and communicated

CAPSTONE FOCUS AREAS

- Deliver End-to-End Working Increment
- Meet Quality Standards
- Collaborate & Communicate
- Manage Risks & Dependencies
- Continuously Integrate & Deploy

TIPS FOR SUCCESS

- Keep the sprint goal specific and achievable.
- Focus on planned scope; avoid overload.
- Inspect and adapt in the Daily Scrum.
- Review and Retrospect to improve.

IDENTIFYING TECHNICAL / DELIVERY RISKS

Unscoreable risks surface as Week 6 thrash and an incomplete Lab 52 defense.

TECHNICAL RISKS

Kafka consumer lag during the live demo.

PostgreSQL migration failure against a shared schema.

JWT misconfiguration -- deny-by-default not enforced.

Contract drift between the UI and the API.

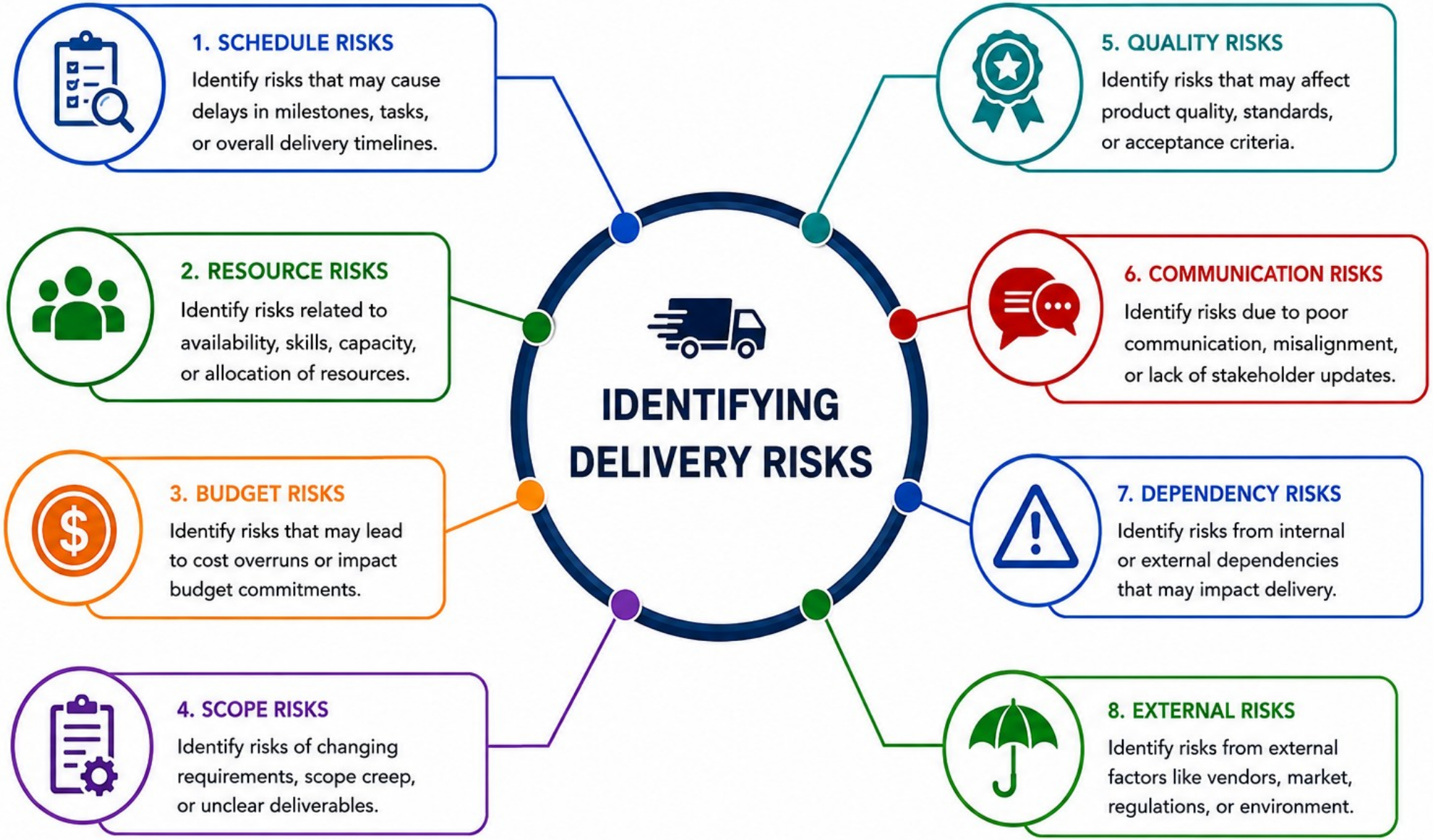
DELIVERY RISKS

- Pipeline secret leak in a committed file.
- Demo environment outage during Lab 52.
- A team member unavailable near a milestone.
- Scope creep past the frozen backlog.

Minimum risk set (Lab 48 requirement)

```
Kafka lag, PostgreSQL migration failure, JWT misconfig,
pipeline secret leak, demo outage, contract drift.
```

KEY TAKEAWAY 'Kafka might fail' with no score or owner cannot be prioritized -- every risk needs both before it counts.



RISK MITIGATION STRATEGIES

A tested control beats 'hope' every time -- mitigations must be as concrete as the risks they answer.

RISK REGISTER COLUMNS

id, risk, likelihood, impact, mitigation, owner, due date, status.

Score = likelihood x impact -- ranks risks, not feelings.

Every risk needs a trigger: what tells you it is happening.

Contingency: what you do if the mitigation itself fails.

WORKED MITIGATIONS

- Kafka lag -> bounded retries + DLT + a lag alert (Lab 49).
- Migration failure -> test the migration against a throwaway DB first.
- JWT misconfig -> negative tests for 401/403 in CI (Lab 51).
- Secret leak -> a pre-commit scan + a .gitignore review.

Risk row (excerpt)

```
Kafka lag | Likely | High | Bounded retry+DLT+alert | Lab 49 owner | Day 2
```

KEY TAKEAWAY 'Hope' is not a mitigation -- replace it with a testable control that has an owner and a due date.

TRY IT YOURSELF -- EXERCISE 5: OUTLINE RISK REGISTER

Reinforces: defining risk-register columns and three starter risks with owners and dates -- unowned risks fail professionalism.

STEPS (notes/lab48-risk-register.md)

Step	What To Do
1 -- Columns	id, risk, likelihood, impact, mitigation, owner, due date, status.
2 -- Starter Risks	Secret leak, Kafka lag during demo, incomplete JWT negative tests.
3 -- Check the Reference	Every risk needs owner + date -- unowned risks fail professionalism.
4 -- Scope	Outline only -- the full register is completed in Lab 48.

Starter risk row

```
Risk: _____   Likelihood: _____   Impact: _____
Mitigation: _____   Owner: _____   Due: _____
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 5 before continuing: exercises/exercise-05-risk-register.md (workspace: examples/module-48-exercises).

ARCHITECTURE DOCUMENTATION

ADRs and RFCs are how architecture decisions survive past the meeting where they were made.

Field	Requirement
Status	Proposed | Accepted | Superseded -- never left blank.
Owners	A named role or team -- never 'someone'.
Alternatives	At least two viable alternatives, or it is not a real decision.
Consequences	Benefits, costs, failure modes, and follow-up work.

REQUIRED ADR TOPICS

PostgreSQL as System of Record

ADR-001 -- justify the persistence choice.

Kafka for Interaction Events

ADR-002 -- justify async messaging over sync-only.

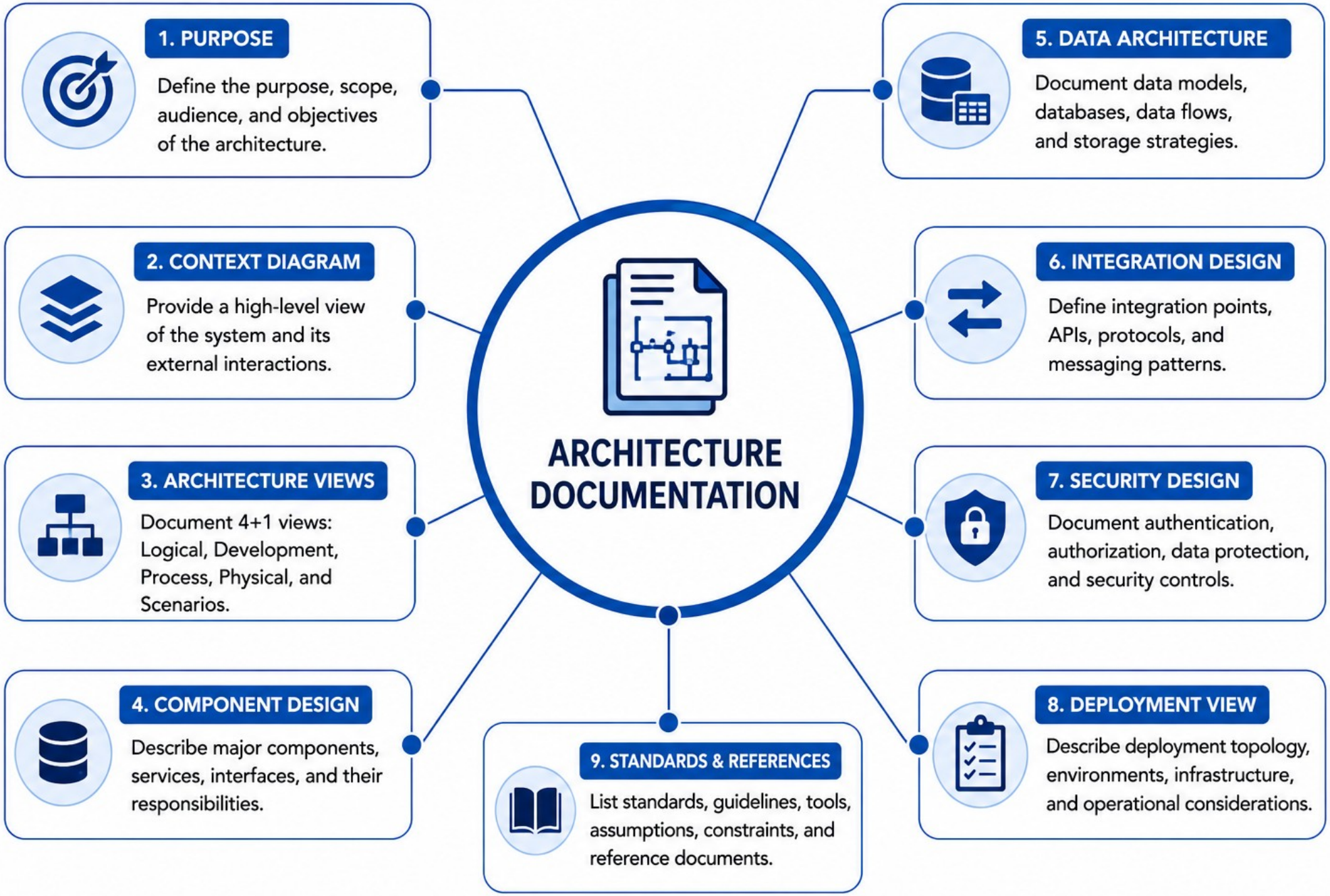
Consistency Strategy

ADR-003 -- after-commit publish vs. outbox, with trade-offs.

JWT / OIDC Authentication

ADR-004 -- resource-server pattern, alternatives considered.

KEY TAKEAWAY Decision-only sticky notes are not ADRs -- expand alternatives and consequences, or the reviewer rejects the decision quality.



TRY IT YOURSELF -- EXERCISE 4: FILL ADR TOPIC TODOS

Reinforces: completing an ADR shortlist with blanks for status and owners across five required topics.

STEPS (notes/lab48-adr-todos.md)

Step	What To Do
1 -- Template	ADR title, status, decision needed by, options A/B, owner -- for each topic.
2 -- Fill Three	Fully fill three ADR stubs; leave two as title-only for Lab 48.
3 -- Consequence Reminder	Add: "Consequences must be written in Lab 48" under each stub.
4 -- No Code	Do not implement any of the decisions yet.

Five required ADR topics

```
API style, Kafka event versioning, authn/z approach,
DB migration strategy, deploy target (k3s).
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 4 before continuing: exercises/exercise-04-adr-todos.md (workspace: examples/module-48-exercises).

PROJECT DOCUMENTATION STANDARDS

Six artifacts, one docs tree -- Labs 49-52 implement and defend against exactly these files.

THE SIX CORE ARTIFACTS

docs/architecture/context.md and container.md.

docs/nfrs.md -- measurable quality targets.

docs/backlog.md -- prioritized vertical stories.

docs/adrs/, docs/risk-register.md, and docs/team-plan.md.

DOCUMENTATION STANDARDS

- Prefer Mermaid/text diagrams in the repo over embedded images only.
- No secrets, real PII, kubeconfigs, or .env files, ever.
- A peer opens the docs alone and restates CAP-12 from them.
- Keep Accepted ADRs -- supersede, never silently delete.

docs/ tree (minimum)

```
docs/architecture/{context,container}.md, docs/nfrs.md,
docs/backlog.md, docs/adrs/, docs/risk-register.md, docs/team-plan.md
```

KEY TAKEAWAY Embedded-image-only diagrams get ignored in review -- commit the Mermaid/text source alongside any picture.

TRY IT YOURSELF -- EXERCISE 6: PLANNING DOCS CHECKLIST

Capstone: build a Pass/Fail checklist mirroring the Week 6 planning portfolio -- the final gate before Lab 48 itself.

STEPS (notes/lab48-docs-checklist.md)

Step	What To Do
1 -- Rows	Context/container, measurable NFRs, backlog+AC, ADRs, risk register.
2 -- Paths	Map each row to a docs/... path from the Week 6 index.
3 -- Branch Note	Note the intended branch name pattern lab/48-crm.
4 -- Scope	Checklist warmup only -- do not claim Lab 48 complete.

Checklist row

```
Artifact: _____   Path: docs/_____   Peer OK? Y/N
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 6 before continuing: exercises/exercise-06-docs-checklist.md (workspace: examples/module-48-exercises). Final gate before Lab 48.

LAB OVERVIEW -- NORTHSTAR CRM EXECUTABLE PLAN

Turn the Northstar CRM brief into an executable architecture and delivery plan that Labs 49-52 implement against.

BUSINESS SCENARIO

- The capstone team must deliver one coherent Customer Management Platform, not five disconnected demonstrations. No Lab 49-52 work counts as in scope unless it maps to a backlog item, an ADR, and a measurable NFR or acceptance criterion.
- You own the planning gate for agent journeys around Amina (CUS-1001, ACTIVE) and Ravi (CUS-1002, PROSPECT to ACTIVE), interaction recording, search, profile, timeline, secure release, and final defense.
- A peer must be able to reproduce the intended Week 6 plan from your docs tree alone, without Slack archaeology.

LEARNING OBJECTIVES

- Produce C4 context and container views with trust boundaries.
- Write measurable NFRs with method, environment, and thresholds.
- Slice a prioritized vertical backlog including CAP-12.
- Record five or more ADRs with alternatives and consequences.
- Plan ownership, critical path, and a scored risk register.
- Keep CUS-1001/CUS-1002/lab-request-001 fixtures consistent throughout.

WHAT YOU BUILD

- customer-management-platform/docs/: architecture/context.md and container.md, nfrs.md, backlog.md, adrs/ (five or more), risk-register.md, and team-plan.md.

KEY TAKEAWAY Fixtures CUS-1001 (Amina Khan) and CUS-1002 (Ravi Singh) are synthetic -- never real customer PII. Duration: about 45 minutes with starters, up to 5-6 hours full path. Advanced capstone difficulty.

LAB TASKS AND DELIVERABLES

lab48-crm -- implementation steps, deliverables, and the evaluation rubric.

#	Implementation Step
1	Clarify product outcome: users, journeys, exclusions, success measures.
2	Model system context: C4 context with protocols and trust boundaries.
3	Design containers and data flow: React, Spring Boot, PostgreSQL, Kafka, IdP.
4	Define domain and contracts: endpoint and event sketches, versioning policy.
5	Write measurable NFRs: latency, availability, security, a11y, retention.
6	Create the prioritized vertical backlog, including CAP-12.
7	Record five or more ADRs: PostgreSQL, Kafka, consistency, JWT, deploy target.
8	Plan delivery and risk: team-plan.md and a scored risk-register.md.

EXPECTED DELIVERABLES

- docs/architecture/context.md and container.md
- docs/nfrs.md
- docs/backlog.md (at least one vertical story including CAP-12)
- docs/adrs/ (five or more ADRs)
- docs/risk-register.md and docs/team-plan.md

RUBRIC (100 MARKS)

- Environment/Structure 10 * Core Impl 30 * Integration/Config 15 * Failure Handling 15 * Verification 10 * Security/Production Awareness 10 * Docs 10

KEY TAKEAWAY Pretty diagrams without trust boundaries or owners lose architecture marks. A backlog that cannot demo the CUS-1001 interaction is incomplete core implementation.

MODULE 48 SUMMARY

In this module, we turned the Northstar CRM brief into an executable plan: architecture, NFRs, backlog, ADRs, and a scored risk register.

KEY CONCEPTS COVERED

C4 Architecture

Context and container views with trust boundaries -- no internals in context.

Measurable NFRs

Threshold + method + environment -- adjectives are banned.

Vertical Backlog

CAP-12 and friends -- ordered by value, risk, dependency, and learning.

ADRs

Status, owners, two or more alternatives, and consequences -- for every real decision.

Scored Risk Register

Likelihood x impact, trigger, mitigation, owner, due date.

Team Plan

Named owners for Labs 49-52 and a visible critical path.

MODULE FLOW RECAP

1. Clarify Outcome
2. Model Architecture
3. Write NFRs & Backlog
4. Record ADRs
5. Plan Risk & Owners

KEY TAKEAWAY Planning is a graded engineering skill: architecture, measurable NFRs, a vertical backlog, and owned risk -- reproducible by a peer without Slack archaeology.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of C4 views, NFRs, vertical backlogs, and ADRs.

1. What must a C4 context diagram avoid including?

A. External systems and people

B. Protocols and trust boundaries

C. Class names and Kafka topic internals

D. The Identity Provider

2. Why is 'the system should be fast' an unacceptable NFR?

A. It is too short

B. It has no threshold, measurement method, or environment

C. It does not mention Kafka

D. It is grammatically incorrect

3. What makes a backlog story 'vertical' rather than 'horizontal'?

A. It only touches the database layer

B. It crosses API, UI, data, and events lightly for one outcome

C. It is written by the Product Owner only

D. It has no acceptance criteria

4. What does an ADR require besides a decision and a status?

A. A single mandatory alternative

B. Nothing else is needed

C. Two or more alternatives and documented consequences

D. A cost estimate in dollars

KEY TAKEAWAY Context stays free of internals; NFRs need threshold + method + environment; backlogs stay vertical; ADRs need real alternatives.

KNOWLEDGE CHECK (2 OF 2)

Four more questions on risk registers, fixtures, traceability, and documentation standards.

5. What two things does every risk-register row require, besides likelihood and impact?

A. A joke and a deadline

B. A mitigation and a named owner

C. A screenshot only

D. A Slack channel link

6. Which fixture ID is reserved for not-found and negative-path testing?

A. CUS-1001

B. CUS-1002

C. CUS-9999

D. lab-request-001

7. No Lab 49-52 work counts as 'in scope' unless it maps to what?

A. A verbal agreement made in standup

B. A backlog item, an ADR, and a measurable NFR or acceptance criterion

C. Whatever looks impressive in the demo

D. The instructor's personal preference

8. What must happen to a deferred backlog story instead of silently dropping it?

A. Delete it from the repository

B. Mark it Explicitly Deferred with an owner and a date

C. Leave it undocumented

D. Move the discussion to a private chat

KEY TAKEAWAY Score every risk with likelihood, impact, mitigation, and owner; keep fixtures consistent; trace every story to backlog + ADR + NFR; defer explicitly, never silently.

"Freeze the plan before you freeze the code -- architecture, ADRs, and a backlog everyone can defend."

MODULE 48 COMPLETE

Capstone Planning and Architecture

You can now produce C4 context and container views, write measurable NFRs, slice a prioritized vertical backlog including CAP-12, record ADRs with real alternatives, and plan a scored risk register -- the frozen plan Labs 49-52 build against.